4) Inductors

Inductors are energy storage devices that operate based on the principle of electromagnetic induction. Solenoids and toroidal coils are typical examples of such components. In the following, we will analyze an ideal toroidal coil to illustrate some fundamental properties shared by all inductors. A toroidal coil is illustrated below (left), along with its cross-sectional view (center) and top-down view (right). The coil is tightly wound with a total of *N* turns. The geometry is defined by the parameters shown in the middle figure.

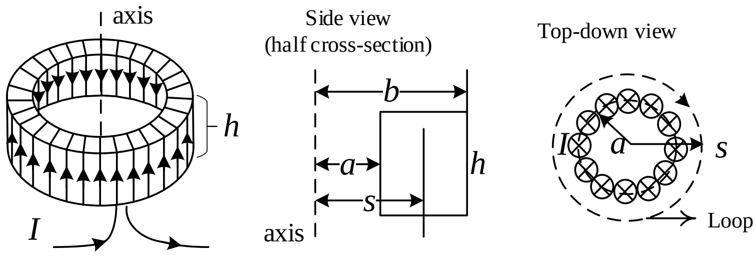

Figure. A toroidal coil (left), its side view (center) and top-down view (right).

As a practice, let us determine the direction of the magnetic field generated by this coil when a steady current *I* is applied.

First, note that the field cannot have a radial component (along the $\hat{s}$ direction). This is because reversing the current would reverse this radial component, yet such a reversal is physically equivalent to flipping the device upside down—an operation that should not change the direction of any radial field component. This contradiction implies that the magnetic field cannot have a radial component.

Second, the device does not generate a magnetic field with a component along the axial direction (i.e., the $\hat{z}$ direction). According to the Biot–Savart law, producing a magnetic field in the $\hat{z}$ direction requires a current with a significant circumferential component (along the $\hat{\varphi}$ direction). However, in a tightly wound toroidal coil, the current primarily flows along the windings, and any circumferential component is negligibly small. Therefore, the resulting magnetic field has no appreciable axial component.

As a result, the only remaining component of the magnetic field is the circumferential one. We now perform an Ampère's loop integral, as illustrated in the above figure (right panel):

$$B \cdot 2\pi s = \mu_0 NI.$$

The circumferential component of the magnetic field depends on the radial distance s from the central axis:

$$B = \frac{\mu_0 NI}{2\pi s}.$$

When the magnetic field varies with time, an EMF is induced in the coil. To evaluate this EMF, we need to calculate the total magnetic flux through the coil, which is given by the flux through a single turn multiplied by N. The flux through a single turn is:

$$\Phi_{B1} = \int \vec{B} \cdot d\vec{a} = \frac{\mu_0 NI}{2\pi} h \int_b^a \frac{1}{s} ds = \frac{\mu_0 NIh}{2\pi} \ln\left(\frac{b}{a}\right),$$

thus the total flux is

$$\Phi_B = N\Phi_{B1} = \frac{\mu_0 N^2 Ih}{2\pi} \ln\left(\frac{b}{a}\right) = LI.$$

Here, $L = \frac{\mu_0}{2\pi} N^2 h \ln\left(\frac{b}{a}\right)$ is the self-inductance, a quantity determined entirely by the geometry of the toroidal coil —specifically, the number of turns N, the inner and outer radii a and b, and the height h. The unit of inductance is henry (H).

5) Energy in Magnetic Fields

In this section, we will study the energy stored in inductors, using which we will generalize to the generic expression of magnetic energy density.

As an energy storage device, an inductor stores energy in the form of a magnetic field. To illustrate this, consider the circuit shown in the figure below, which consists of a battery (with EMF $\mathcal{E}_0$), an inductor L, a resistor R, and a switch that is initially open. These components are connected in series.

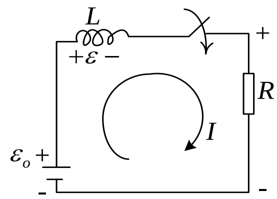

Figure. A circuit consisting of an inductor.

When the switch is closed, a current will flow. The inductor is made of perfect conductor thus it has a resistance of zero, so the current will finally settle at $\mathcal{E}_0/R$. However, the current will not immediately reach this final value, since an abrupt jump in the current would lead to $\frac{d\Phi_B}{dt}$ approaching infinity, thus an infinitely large EMF. In fact, the voltage drop on the inductor V_{ind}, which can be derived from the EMF $\mathcal{E}_{\text{ind}}$, is $L\frac{dI}{dt}$ (we will discuss more on this issue later). As a result, the total voltage drop on the resistor is:

$$V_R = \mathcal{E}_0 - V_{\text{ind}} = \mathcal{E}_0 - L\frac{dI}{dt} = IR.$$

The above ordinary differential equation can be solved directly for the current:

$$I = \frac{\mathcal{E}_0}{R}\left(1 - e^{-\frac{R}{L}t}\right).$$

As expected, the current will gradually increase until it reaches the maximum value of $\frac{\mathcal{E}_0}{R}$, as is shown in the figure below.

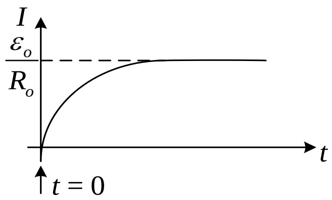

Figure. Transient of the current after the switch is closed.

At any instance, the current through the inductor is *I*, while the EMF (voltage) is $-L\frac{dI}{dt}$ (the negative sign means the EMF opposes the increase of the current), thus the instantaneous power is

$$P(t) = \mathcal{E}I = -LI\frac{dI}{dt},$$

and the total work done by the EMF is:

$$\int P(t)dt = -L\int_0^{I_0} IdI = -\frac{1}{2}LI_0^2 \ ,$$

where $I_0 = \frac{\mathcal{E}_0}{R}$ is the final current. Here, the EMF does negative work to convert the electric energy of the battery into electromagnetic energy stored in the magnetic field of the inductor:

$$W_B = \frac{1}{2}LI_0^2 \ = \frac{1}{2}(LI_0)\cdot I_0$$
$$= \frac{1}{2}\Phi_B \cdot I_0 = \frac{1}{2}I_0 \cdot \int \vec{B}\cdot d\vec{a} = \frac{1}{2}I_0 \cdot \int (\nabla \times \vec{A})\cdot d\vec{a}$$
$$= \frac{1}{2}I_0 \oint \vec{A}\cdot d\vec{\ell} = \frac{1}{2}\oint \vec{A}\cdot \vec{I}_0 d\ell$$

The above integral is around the perimeter of the circuit (including the inductor coil), and assume the electric wire is infinitely thin. In the real case of finite wire thickness, we generalize the above expression to a volume integral:

$$W_B \ = \frac{1}{2}\int_V \vec{A}\cdot\vec{J}_0 dV = \frac{1}{2\mu_0}\int_V \vec{A}\cdot(\nabla\times\vec{B})dV$$

Taking advantage of the identity $\nabla\cdot\left(\vec{A}\times\vec{B}\right) = \vec{B}\cdot\left(\nabla\times\vec{A}\right) - \vec{A}\cdot\left(\nabla\times\vec{B}\right)$, we get:

$$W_B = \frac{1}{2\mu_0}\int_V \left[\vec{B}\cdot\underbrace{(\nabla\times\vec{A})}_{=\vec{B}} - \nabla\cdot(\vec{A}\times\vec{B})\right] dV$$

$$= \frac{1}{2\mu_0}\int_V B^2 dV - \underbrace{\frac{1}{2\mu_0}\oint_S (\vec{A}\times\vec{B})\cdot d\vec{a}}_{\text{vanishes when } S\to\infty}$$

Finally, we conclude that the energy stored in the magnetic field is

$$W_B = \int_{\text{all space}} \frac{1}{2\mu_0} B^2 dV,$$

while the magnetic energy density is

$$\boxed{w_B = \frac{1}{2\mu_0} B^2}.$$

Note:

① The mathematical form of the electric energy density, $w_e = \frac{\varepsilon_0}{2}E^2$, completely mirrors that of the magnetic energy density.

② Let's check the formula using the toroidal coil, for which $B = \frac{\mu_0 NI}{2\pi s}$, $dV = sdsd\varphi dh$

$$W_B = \frac{1}{2\mu_0}\int B^2 dV = \frac{1}{2\mu_0}\int_a^b \left(\frac{\mu_0^2 N^2 {I_0}^2}{4\pi^2 s^2}\right) sds \int_0^h dh' \int_0^{2\pi} d\varphi$$

$$= \frac{\mu_0 N^2 {I_0}^2 h}{4\pi} \ln\left(\frac{b}{a}\right) = \frac{1}{2}\underbrace{\frac{\mu_0 N^2}{2\pi}\left(h \ln\frac{b}{a}\right)}_{L}{I_0}^2$$

$$= \frac{1}{2} L {I_0}^2$$

2. Displacement current

We recall Ampere's Law in its integral form:

$$\oint_L \vec{B}\cdot d\vec{\ell} = \mu_0 I_{enc},$$

where $\oint_L \vec{B}\cdot d\vec{\ell}$ is the line integral of the magnetic field around a closed loop L, and I_{enc} is the enclose current passing through the loop.

Now, let's consider the simplest scenario to investigate the potential limitations of Ampere's Law, as shown below.

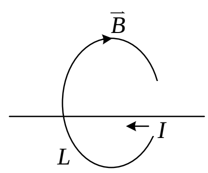

Figure. A simple situation to apply Ampere's Law.

In the situation where we perform a loop integral around a straight wire carrying a steady current, Ampere's Law holds perfectly.

However, when dealing with a time-varying current, an awkward situation arises. Ampere's Law, in its original form, only accounts for the magnetic field generated by steady currents. When the current is changing with time, the situation becomes more complex. Consider a plate capacitor being charged by a constant current, as illustrated in the figure below. While a steady current flows through the wire, on the capacitor itself, the charge density ρ is changing with time, i.e., $\frac{\partial}{\partial t}\rho \neq 0$. This time variation in charge density violates the condition for a steady current.

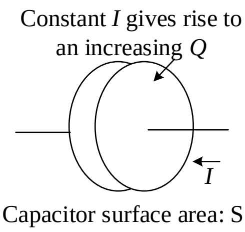

Figure. A plate capacitor being charged by a constant current.

Under this new setup, if we apply Ampere's Law by choosing a flat surface bounded by the loop (as shown in the left panel of the figure below), the result is consistent and shows no violation of the law.

$$\oint_L \vec{B} \cdot d\vec{\ell} = \mu_0 \int_{S_1} \vec{J} \cdot d\vec{a} = \mu_0 I_{\text{enc}}.$$

Recall that we are free to choose the integration surface. Now, if we choose a curved surface S_2 that passes through the gap between the two metal plates, as shown in the right panel of the figure below, we encounter a different situation. In this case, the surface encloses no current at all, which leads to the following result:

$$\oint_L \vec{B} \cdot d\vec{\ell} = \mu_0 \int_{S_2} \vec{J} \cdot d\vec{a} = 0.$$

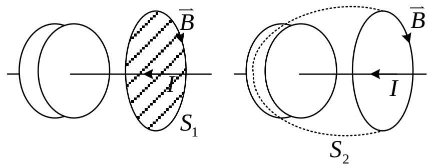

Figure. Two different types of integration surfaces.

Apparently, integrating over S1 and S2 yields completely different results. To resolve this discrepancy, we recognize that the time-varying charge density on the two plates results in a time-varying electric field. Since a time-varying magnetic field induces an electric field, it is reasonable to assume that a time-varying electric field similarly produces a magnetic field. This was, in fact, Maxwell's key insight. He resolved the issue by introducing the displacement current, ensuring the consistency of Ampere's Law in dynamic situations.

To understand the concept of displacement current more quantitatively, we can refine the idea as follows: A constant current I gives rise to an increasing charge amount Q on one of the plates (for the other plate, the charge amount is $-Q$), and we know that the current is related to the charge by

$$\frac{dQ}{dt} = I.$$

As the charge accumulates, the electric field between the two capacitor plates changes. This is because the electric field between the plates of a parallel-plate capacitor is given by:

$$E = \frac{\sigma}{\varepsilon_0},$$

where σ is the charge density which is assumed to be uniform. From the above equation, we obtain:

$$ES = \frac{\sigma S}{\varepsilon_0} = \frac{Q}{\varepsilon_0},$$

where S is the surface area of the plate and

$$S \cdot \frac{\partial E}{\partial t} = \frac{1}{\varepsilon_0}\frac{dQ}{dt} = \frac{I}{\varepsilon_0},$$

Consequently,

$$\varepsilon_0 \cdot \frac{\partial E}{\partial t} = \frac{I}{S} = \mathcal{J}_d,$$

where $\mathcal{J}_d$ is an auxiliary quantity that has the units of current density.

This current is known as the displacement current. By substituting $\mathcal{J}_d$ for $\mathcal{J}$ in the integration over S_2, we resolve the issue:

$$\mu_0 \int_{S_1} \vec{\mathcal{J}}_d \cdot d\vec{a} = \mu_0 \int_{S_2} \varepsilon_0 \cdot \frac{\partial \vec{E}}{\partial t} \cdot d\vec{a} = \mu_0 \int \frac{I}{S} \cdot ds = \mu_0 I.$$

In summary, through the study of this particular case, we learn that:

$$\oint_L \vec{B} \cdot d\vec{\ell} = \mu_0 \int_{\substack{\text{any surface} \\ \text{attached to } L}} \left(\underbrace{\vec{\mathcal{J}}}_{=0 \text{ in gap}} + \underbrace{\varepsilon_0 \frac{\partial \vec{E}}{\partial t}}_{=0 \text{ across wire}} \right) \cdot d\vec{a} = \int_{\substack{\text{any surface} \\ \text{attached to } L}} \nabla \times \vec{B} \cdot d\vec{a}$$

This equation holds for any surface attached to the loop L, this is possible only when

$$\boxed{\underbrace{\nabla \times \vec{B} = \mu_0 \vec{J}}_{\text{Magnetostatic}} + \underbrace{\mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}}_{\text{Maxwell's addition}}}$$

The equation above captures all aspects of the phenomena related to the curl of the magnetic field, and concludes our exploration of Maxwell's four equations.

We can derive Maxwell's addition to the original Ampère's Law from a purely mathematical perspective. The incompleteness of Ampère's Law becomes evident when we apply the divergence operator to both sides of the equation

$$\nabla \times \vec{B} = \mu_0 \vec{J}.$$

The left-hand side is always zero:

$$\nabla \cdot (\nabla \times \vec{B}) \equiv 0,$$

because the divergence of a curl is identically zero. However, the right-hand side is zero only under the condition of a steady current:

$$\mu_0 (\nabla \cdot \vec{J}) = 0.$$

In the general case, the continuity equation tells us:

$$\nabla \cdot \vec{J} = -\frac{\partial}{\partial t} \rho,$$

which is not zero when the charge distribution is time-dependent. This inconsistency reveals the limitation of Ampere's original law and motivates the need for a correction—namely, the introduction of the displacement current.

To resolve the problem, we begin by recalling Gauss's Law:

$$\rho = \varepsilon_0 \nabla \cdot \vec{E}.$$

Substituting this into the continuity equation, we obtain:

$$\nabla \cdot \vec{J} = -\frac{\partial}{\partial t} \rho = -\frac{\partial}{\partial t} (\varepsilon_0 \nabla \cdot \vec{E}).$$

This expression suggests that if we modify Ampere's Law by adding an additional term $\mu_0 \varepsilon_0 \frac{\partial}{\partial t} \vec{E}$, then the inconsistency is resolved. To wit,

$$\boxed{\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial}{\partial t} \vec{E}}.$$

To verify that the modified Ampere's Law resolves the previous issue, we apply the divergence operator to both sides of the new equation. This yields:

$$\underbrace{\nabla \cdot (\nabla \times \vec{B})}_{\substack{\text{Divergence of a curl} \\ \text{is always zero}}} = \mu_0 \left(\nabla \cdot \vec{J} + \frac{\partial}{\partial t} (\varepsilon_0 \nabla \cdot \vec{E}) \right) = \mu_0 \underbrace{\left(\nabla \cdot \vec{J} + \frac{\partial}{\partial t} \rho \right)}_{\substack{=0 \text{ due to continuity} \\ \text{equation}}} \equiv 0.$$

Note:

① In the absence of $\vec{J}$, we have $\nabla \times \vec{B} = \mu_0 \varepsilon_0 \partial_t \vec{E}$, which means that a time-varying electric field generates a rotational magnetic field. This mirrors the Maxwell–Faraday equation, where a time-varying magnetic field generates a rotational electric field.

② The displacement current is often overshadowed by the presence of $\vec{J}$. As a result, it is less prominent and harder to detect experimentally, unlike Faraday's induction, which is more readily observed—partly because there is no known equivalent of magnetic current to cancel or obscure its effects.

③ Unlike Lenz's Law, which features a negative sign indicating opposition (consistent with energy conservation), the displacement current term in Maxwell's equations comes with a positive sign. This sign is not arbitrary—it arises directly from the continuity equation, ensuring charge conservation, and is essential for the formation and propagation of electromagnetic waves.

3. Maxwell's Equations

Bravo! We have now completed our study of the four Maxwell's equations, which are summarized below:

Name	Differential form	Integral form
Gauss's Law	$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$	$\oint_S \vec{E} \cdot d\vec{a} = \frac{Q_{\text{enc}}}{\varepsilon_0}$
Gauss's Law for B	$\nabla \cdot \vec{B} = 0$	$\oint_S \vec{B} \cdot d\vec{a} = 0$
Maxwell-Faraday equation	$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	$\oint_L \vec{E} \cdot d\vec{\ell} = -\frac{d\Phi_B}{dt}$
Ampere's Law with Maxwell's addition	$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial}{\partial t} \vec{E}$	$\oint_L \vec{B} \cdot d\vec{\ell} = \mu_0 \left(I_{\text{enc}} + \varepsilon_0 \frac{d\Phi_E}{dt} \right)$

Maxwell's equations inherently requires local charge conservation, as described by the continuity equation:

$$\nabla \cdot \vec{J} = -\frac{\partial}{\partial t} \rho,$$

as we discussed above (by taking the divergence of Ampere-Maxwell's law). However, Maxwell's equations only dictate how charges and currents produces fields, but do not specify how these fields act back on charges. The reaction of electromagnetic fields on charges is described by the Lorentz force law:

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B}).$$

Maxwell's equations, as shown above, are general and apply both in vacuum and in matter. When matter is involved, the total charge density ρ includes contributions from both free charges ρ_f and bound charges ρ_b. Similarly, the total current density $\vec{J}$ includes both free current $\vec{J}_f$ and bound current $\vec{J}_b$. The bound charge and current densities, ρ_b and $\vec{J}_b$ arise from the microscopic polarization and magnetization of the material. They are induced quantities, dependent on the material's internal response to the applied fields, and are generally not known *a priori*.

To handle this complexity, we introduce two new auxiliary fields: the electric displacement field $\vec{D}$ and the magnetic vector $\vec{H}$, which allow us to absorb the effects of bound charges and currents into the material properties. This leads to the macroscopic form of Maxwell's equations, where only the free charges and currents appear explicitly, and the material response is described through constitutive relations.

To derive the macroscopic form of Maxwell's equations, we already have all the necessary ingredients—except for one missing piece of the puzzle: the polarization current. To understand this concept more clearly, let us begin by examining a familiar scenario, as illustrated in the figure below.

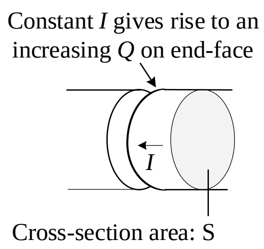

Figure. A special capacitor formed by a gap cut in a conducting wire.

Similar to the charging capacitor discussed earlier, consider a capacitor formed by introducing a gap (perpendicular to the axis) in a conducting wire of finite diameter. Let the cross-sectional area of the wire be *S*. As the capacitor is charged with a constant current *I*, charge accumulates on the end faces of the gap. The rate of charge accumulation is given by:

$$\frac{dQ}{dt} = I.$$

Dividing both sides by the area *S*, we obtain:

$$\frac{1}{S}\frac{dQ}{dt} = \frac{\partial \sigma}{\partial t} = \frac{I}{S} = \vec{J} \cdot \hat{n}.$$

In words, a time-varying surface charge density σ is sustained by a current density $\mathcal{J}$. This highlights the connection between current in the wire and the dynamic behavior of charge density.

Now, suppose we have a thin slab of material with a time-varying polarization $\vec{P}(t)$. As we've learned, this induces a surface bound charge density given by $\sigma_b(t) = \vec{P}(t) \cdot \hat{n}$, which also varies with time. From our earlier conclusion that a time-varying surface charge density σ is sustained by a current density $\mathcal{J}$, such that $\frac{\partial \sigma}{\partial t} = \vec{\mathcal{J}} \cdot \hat{n}$, we can apply this to the bound charge density. Therefore, a time-varying surface bound charge density is associated with a special type of current:

$$\frac{\partial \sigma_b}{\partial t} = \frac{\partial \vec{P}}{\partial t} \cdot \hat{n} = \vec{\mathcal{J}}_p \cdot \hat{n},$$

This leads to the expression for the polarization current:

$$\boxed{\vec{\mathcal{J}}_p = \frac{\partial \vec{P}}{\partial t}}$$

As expected, a time-varying polarization must be accompanied by the movement of bound charges, which in turn generates a current. Note that the polarization current is generated by the movement of bound charges, but it is completely distinct from bound currents, which are due to the magnetization of the material. From the above definition, we can easily derive the continuity equation for the bound charge:

$$\nabla \cdot \vec{\mathcal{J}}_p = \frac{\partial \nabla \cdot \vec{P}}{\partial t} = -\frac{\partial \rho_b}{\partial t}.$$

Now we are ready to derive the macroscopic version of Maxwell's equations. Let us review the key properties of bound charges and currents:

$$\rho_b = -\nabla \cdot \vec{P},$$

$$\vec{\mathcal{J}}_b = \nabla \times \vec{M},$$

and the constitutive relations:

$$\vec{D} = \varepsilon_0 \vec{E} + \vec{P},$$

$$\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M}.$$

We note that there are two types of charges, namely, free charge and bound charge:

$$\rho = \rho_f + \rho_b = \rho_f - \nabla \cdot \vec{P},$$

and three types of currents, namely, free current, bound current and polarization current:

$$\vec{\mathcal{J}} = \vec{\mathcal{J}}_f + \vec{\mathcal{J}}_b + \vec{\mathcal{J}}_p = \vec{\mathcal{J}}_f + \nabla \times \vec{M} + \frac{\partial}{\partial t}\vec{P}.$$

These relations can be used to derive the macroscopic Maxwell's equations—in terms of $\vec{D}$ and $\vec{H}$—starting from the original equations.

From Gauss's Law:

$$\nabla \cdot \vec{E} = \frac{1}{\varepsilon_0}(\rho_f - \nabla \cdot \vec{P}) \Rightarrow \nabla \cdot \underbrace{(\varepsilon_0 \vec{E} + \vec{P})}_{\vec{D}} = \rho_f \Rightarrow$$

$$\boxed{\nabla \cdot \vec{D} = \rho_f}$$

From Ampere's Law with Maxwell's addition:

$$\nabla \times \vec{B} = \mu_0 \left(\vec{J}_f + \nabla \times \vec{M} + \frac{\partial}{\partial t}\vec{P}\right) + \mu_0 \varepsilon_0 \frac{\partial}{\partial t}\vec{E} \qquad \Rightarrow$$

$$\nabla \times \underbrace{\left(\frac{\vec{B}}{\mu_0} - \vec{M}\right)}_{\vec{H}} = \vec{J}_f + \underbrace{\frac{\partial}{\partial t} \underbrace{(\varepsilon_0 \vec{E} + \vec{P})}_{\vec{D} \to \text{displacement}}}_{\partial_t \vec{D}\text{: displacement current}} \qquad \Rightarrow$$

$$\boxed{\nabla \times \vec{H} = \vec{J}_f + \frac{\partial}{\partial t}\vec{D}}$$

while the rest two equations, $\nabla \cdot \vec{B} = 0$, $\nabla \times \vec{E} = -\partial_t \vec{B}$, remain unchanged.

4. Boundary conditions

We employ the integral form of Maxwell's equations to derive the boundary conditions. In the case of time-dependent electric and magnetic fields, the discontinuities of the field components across boundaries, which we observed in static cases, remain unchanged. This is because the only differences in the dynamic case come from the following two equations:

$$\oint_L \vec{E} \cdot d\vec{\ell} = -\frac{d\Phi_B}{dt}$$

and

$$\oint_L \vec{B} \cdot d\vec{\ell} = \mu_0 \left(I_{\text{enc}} + \varepsilon_0 \frac{d\Phi_E}{dt}\right).$$

In static cases, the terms $\frac{d\Phi_B}{dt}$ and $\frac{d\Phi_E}{dt}$ are zero. However, when deriving the boundary conditions using these equations, we reduce the height of the integration loop to zero. As a result, the flux terms Φ_B and Φ_E, as well as $\frac{d\Phi_B}{dt}$ and $\frac{d\Phi_E}{dt}$, also vanish, leading to the same conclusions as we derived in the static case.

5. Poynting's Theorem

Up to now, we have learned how charges and currents generate fields, and how fields exert forces back onto charges. For a deeper understanding of physics, it is important to analyze what happens to electromagnetic energy. This section provides a framework for understanding how electromagnetic energy evolves within a volume *V* bounded by a closed surface *S*.